Brody Fiorito

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EDUCATION

University of Pittsburgh **Expected: May 2029**
B.S. Electrical Engineering Pittsburgh, PA

- GPA: 3.90/4.00
- Relevant Coursework: Linear Circuits & Systems, Digital Circuits & Systems, ECE Problem Solving with C++

EXPERIENCE

Panther Racing - Pitt Formula SAE Electric
Autonomous Subteam Lead **Jul 2026 – Present**

- Building the perception and control stack in C++/ROS 2 Humble — 14 nodes including a VLP-16 lidar cone detector, CAN bridge, and watchdog/state machine — deployed to Jetson via containerized builds
- Relaunched a subteam dormant for two years and managed a $6,000 driverless hardware budget

Low Voltage Engineer **Aug 2025 – Jul 2026**

- Designed a CAN expansion board (STM32C092, TCAN3404-Q1, TPS542025) and wrote HAL firmware converting 8 analog and 2 digital sensor inputs to scheduled CAN frames
- Designed the PR-037 dashboard wiring harness in RapidHarness across 12 connectors, routing CAN, shutdown circuit, power, and signal lines with pinout and build documentation

Teaching Assistant - ENGR 0711: Introduction to Engineering Analysis **Aug 2026 – Dec 2026**

- Support 60 students in C and MATLAB across weekly lab sections and office hours; grade assignments and provide written feedback on code

PROJECTS

LiTel - Vehicle Telemetry Link | Panther Racing **Jul 2026 – Present**

- Designed a 4-layer mixed-signal PCB around an STM32H533 that ingests classic CAN at 1 Mbit/s and relays frames over a 900 MHz RFD900ux link to a pit-side ground station; wide-Vin automotive front end (TVS/fuse/P-FET protection, TPS54360-Q1 buck) operating from the vehicle LV bus
- Wrote the embedded C firmware stack — interrupt-driven CAN RX with full-FIFO drain, ring buffer, and CRC-checked COBS framing — targeting a sub-200 ms end-to-end latency budget, 5x tighter than the previous system

Lissajous Signal Generator - ESP32 Mixed-Signal PCB **Mar 2026 – May 2026**

- Designed a mixed-signal PCB generating phase- and frequency-controlled stereo sine waves, streamed over I^2S to a PCM5102A DAC for Lissajous figure visualization on an oscilloscope
- Developed FreeRTOS firmware in Embedded C with a dual-mode state machine for waveform generation and USB audio passthrough, using ring-buffered audio and ADC conditioning with hysteresis and EMA filtering
- Fabricated at JLCPCB and hand-assembled the board, functional on first power-up; verified rails, I^2S timing, and DAC output on an oscilloscope

TECHNICAL SKILLS

- **Programming & Scripting:** Embedded C, C++, MATLAB, VHDL, Python, Assembly
- **Software & Tools:** Altium Designer, ROS 2 (Humble), Docker, Linux, Git, RapidHarness, LTSpice
- **Hardware & Embedded Platforms:** ESP32 (ESP-IDF, FreeRTOS), STM32 (HAL, CubeMX), CAN bus (pCAN, MoTeC M1), SPI, I^2C, I^2S, UART, ADC, USB